

NOTE: This disposition is nonprecedential.

**United States Court of Appeals
for the Federal Circuit**

APPLE INC.,
Appellant

v.

SMART MOBILE TECHNOLOGIES LLC,
Appellee

2024-1624

Appeal from the United States Patent and Trademark
Office, Patent Trial and Appeal Board in No. IPR2022-
01222.

Decided: December 19, 2025

ANGELA M. OLIVER, Haynes and Boone, LLP, Washing-
ton, DC, argued for appellant. Also represented by
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Before DYK, CHEN, and STOLL, *Circuit Judges*.

CHEN, *Circuit Judge*.

Apple Inc. (Apple) timely appeals the final written decision of the Patent Trial and Appeal Board (Board) ruling that claims 1–12, 14, 19, and 24 of U.S. Patent No. 8,982,863 ('863 patent) had not been shown to be unpatentable as obvious over U.S. Patent No. 5,970,059 (Ahopelto) in combination with other references. *Apple Inc. v. Smart Mobile Techs. LLC*, No. IPR2022-01222, 2024 WL 306227, at *18 (P.T.A.B. Jan. 26, 2024) (*Final Written Decision*). Because the Board adopted the parties' agreed-upon claim construction and substantial evidence supports its conclusion, we *affirm*.

BACKGROUND

The '863 patent relates to wireless enhancements to IP based cellular telephones/mobile wireless devices (CT/MD). '863 patent, col. 1 ll. 43–45. The '863 patent has two independent claims 1 and 14, each of which recites a system comprising a “server” and a “network switch box.” Claim 1 is representative and recites:

1. A system for controlling network Internet Protocol (IP) based wireless devices, IP based cellular phones, networks or network switches by servers comprising:

an IP enabled wireless device including a portable device or a cellular phone, said IP enabled wireless device comprising a plurality of antennas and ports, wherein the IP enabled wireless device is configured for voice and data communication and comprises a plurality of transmit and receive units;

a first *server* connected to at least one internet protocol enabled network, said

server configured with a controller in communication with a plurality of network devices; and

a *network switch box*, wherein the network switch box is configured with a plurality of ports, wherein the network switch box is connected to at least two networks, wherein the network switch box is configured to transmit and receive one or more data packets between the at least two networks.

'863 patent at claim 1 (emphases added).

Apple filed an *inter partes* review petition challenging claims 1–12, 14, 19, and 24 of the '863 patent under 35 U.S.C. § 103 over Ahopelto in combination with other references. The petition relied on Ahopelto for the “server” and the “network switch box,” arguing that Ahopelto’s GPRS gateway support node (GGSN) has two functionalities: a server functionality disclosing the “server” and a separate routing functionality disclosing the “network switch box.” Smart Mobile Technologies LLC (Smart Mobile) responded, arguing that Apple relied on an imagined distinction between the two functionalities. J.A. 263 (“[T]he alleged ‘server functionality’ and ‘routing functionality’ are *not* separate and distinct.”), 357–358.

The Board agreed with Smart Mobile, finding that “[t]he parties agree that the claim requires a server and a network switch box as separate and distinct components” and determining that none of the challenged claims were unpatentable because Apple failed to prove that Ahopelto’s GGSN contains a server functionality that is separate and distinct from its routing functionality. *Final Written Decision*, 2024 WL 306227, at *10. Apple appeals, arguing that (1) the Board incorrectly construed the claims to require a “server” that is logically separate and distinct from a “network switch box” and (2) the Board’s obviousness

determination is not supported by substantial evidence. We have jurisdiction under 28 U.S.C. § 1295(a)(4)(A).

DISCUSSION

I. Claim Construction

Apple contends that the Board erred by assuming that the parties agreed upon the claim construction requiring the claim term “server” be logically separate and distinct from the claim term “network switch box.” Apple asserts that it did not agree to that construction. *See* Appellant Br. 29–30 (explaining that the petition suggested both claim terms be given their plain and ordinary meaning).

The record before us indicates otherwise. Apple’s unpatentability case, from the petition to an oral hearing, was all along premised on understanding the claims to require two logically separate and distinct functions. For example, Apple’s petition stated, “routing functionality [is] *in addition to and distinct* from the server functionality for controlling packet forwarding.” J.A. 144 (emphasis added); *see also* J.A. 145 (“The GGSN routing functionality would be implemented as a logical entity on a shared platform . . .”). Apple’s reply made its position clear: “the question is whether a POSITA would have found the claimed ‘*server*’ and ‘*network switch box*’ obvious from Ahopelto’s *logically distinct* GGSN function teachings.” J.A. 315 (third emphasis added); *see also* J.A. 311 (“The evidence demonstrates it was known that GGSNs (like those systems taught in Ahopelto) include logical components for different functions, such as ‘*server*’ and ‘*network switch box*’ functionalities.”). Apple’s expert Dr. Jensen operated under the same understanding of the asserted unpatentability theory. *See* J.A. 730 ¶ 121, J.A. 731 ¶ 123 (explaining that a skilled artisan would have recognized that GGSN’s different functionalities “would be implemented as different logical (e.g., software) entities on a shared platform.”). At the hearing, in response to the Board’s question about “the connection between what is a server and what is a

network switch box,” Apple responded that “one entity[] can have multiple different logical elements within it.” J.A. 401–02. Smart Mobile agreed that the “server” and the “network switch box” need not be physically separate, but asserted that Apple still must establish the two as distinct components. J.A. 262–63, 357–58.

Given this record, it was proper for the Board to conclude that “[t]he parties agree that the claim requires a server and a network switch box as separate and distinct components.” *Final Written Decision*, 2024 WL 306227, at *10. The Board acknowledged Smart Mobile’s concession that two claim elements need not be physically separate, *id.* at *8, and simply adopted the parties’ understanding that the claims require logically separate and distinct server and routing functionalities, *id.* at *10 (citing Apple’s “logically distinct” language). Because the parties agreed on this claim construction throughout the IPR,¹ the Board did not err in evaluating Apple’s patentability challenge based on Apple’s understanding of what the claim requires.

Apple argues in the alternative that the Board’s construction is unworkable in the context of software inventions. *See* Appellant Br. 32–33 (questioning “whether such a [logical separate-and-distinctness] requirement even makes sense in the context of device functions (e.g., software functions)”); *see also id.* at 36. In Apple’s view, the Board’s construction presents a difficult box-drawing

¹ For this reason, Apple’s reliance on *Bot M8* is misplaced. *See* Appellant Br. 31, 36 (citing *Bot M8 LLC v. Sony Interactive Ent. LLC*, No. 2022-1569, 2023 WL 5606978 (Fed. Cir. Aug. 30, 2023) (non-precedential)). Here, unlike *Bot M8*, Apple consistently advanced a view to the Board that the claims require logically separate and distinct server and routing functions. *See* J.A. 144–45, 311, 315, 401–02, 730–31.

problem—“how does one identify where one software functionality ends and another begins?” *Id.* at 33.

Apple forfeited this argument by not presenting it to the Board. “We have long held that a party forfeits an argument that it fails to present to the Board because that failure deprives this court of the benefit of the Board’s informed judgment.” *Voice Tech Corp. v. Unified Pats., LLC*, 110 F.4th 1331, 1340 (Fed. Cir. 2024) (citing *In re NuVasive, Inc.*, 842 F.3d 1376, 1380 (Fed. Cir. 2016)).

Instead, Apple’s obviousness theory was built on an implicit construction of the “server” and the “network switch box” as defining logically separate and distinct functions. Apple demonstrated this understanding in three ways. First, Apple did not ask for claim construction in the petition, while presuming that the claims require a logical separate-and-distinctness. J.A. 144–45. Second, Apple did not raise in the reply, before the Board, its current argument against separate-and-distinctness, even after Smart Mobile argued that the petition did not demonstrate logically separate and distinct functions. J.A. 262–68. Rather, Apple asserted that “the question is whether a POSITA would have found the claimed ‘*server*’ and ‘*network switch box*’ obvious from Ahopelto’s logically distinct GGSN function teachings.” J.A. 315. Third, Apple did not file a supplemental expert declaration to support its current theory or to rebut Smart Mobile’s expert. Accordingly, we find that Apple forfeited its unworkability argument.

Because we find that the Board correctly adopted the parties’ agreed-upon claim construction, we need not address Apple’s argument that as a matter of claim construction, the claims do not require the logical separate-and-distinctness. *See* Appellant Br. 37–43.

II. Obviousness Determination

We next address Apple’s challenge to the Board’s obviousness determination. “Obviousness is a question of law

based on underlying facts.” *Virtek Vision Int’l ULC v. Assembly Guidance Sys., Inc.*, 97 F.4th 882, 886 (Fed. Cir. 2024). “We review the Board’s ultimate determination of obviousness de novo and its underlying facts for substantial evidence.” *Id.* “Substantial evidence review asks ‘whether a reasonable fact finder could have arrived at the agency’s decision. . . .’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1366 (Fed. Cir. 2016) (quoting *In re Gartside*, 203 F.3d 1305, 1312 (Fed. Cir. 2000)).

Apple contends that even if the claims require logical separate-and-distinctness, Ahopelto’s GGSN discloses two functionalities—the server functionality and the routing functionality—that are separate and distinct from each other. *See* Appellant Br. 43. Apple offers two arguments for why the Board’s obviousness determination cannot stand. First, Apple contends that the Board applied an erroneous standard for disclosure by looking at whether Ahopelto recites claim language verbatim or presents an architecture diagram showing various functionalities. *Id.* at 44–46. Second, Apple contends that the Board overlooked and conflated evidence when discounting Dr. Jensen’s testimony. *See* Appellant Br. 47–51.

We find both arguments unpersuasive. Contrary to Apple’s characterization, the Board’s obviousness determination was not based on a mere fact that Ahopelto did not recite claim terms verbatim or present an architecture diagram delineating GGSN’s functions. Rather, the Board engaged in a three-tiered analysis, by first looking at disclosures in Ahopelto, second in testimonial evidence, and third in additional cited evidence. *Final Written Decision*, 2024 WL 306227, at *10–13. Apple’s unpatentability theory was based on Ahopelto teaching both the “server” and the “network switch box.” J.A. 139–40, 144–46. Under the first step of its analysis, the Board found that “Ahopelto does not describe expressly that its GGSN is ‘a server’ and a separate and distinct ‘network switch box.’” *Final*

Written Decision, 2024 WL 306227, at *10. Then, the Board looked at the testimonial evidence and additional cited evidence and concluded that Apple had not met its burden to show that a skilled artisan “would have understood that Ahopelto’s GGSN includes a server functionality and a separate and distinct routing functionality.” *Id.* at *12.

The Board’s findings are supported by substantial evidence. For example, nothing in Ahopelto’s specification discusses GGSN having separate and distinct functionalities for determining how to forward a packet based on the packet’s protocol type (Apple’s mapping of a “server”) and for connecting data packet networks to each other (Apple’s mapping of a “network switch box”).² *Final Written Decision*, 2024 WL 306227, at *10. Ahopelto instead describes that “[t]he routing procedure” encompasses the server functionality, namely the step of “forward[ing] the [data] packet” upon “determining the protocol type of the

² The parties agree that the server functionality relates to how to forward a packet. *See* Appellant Br. 17, 56–57; *see* Appellee Br. 10, 52. However, the parties dispute the meaning of routing functionality. Apple argues that it pertains to where to forward a packet, while Smart Mobile argues that it concerns connecting data networks. *See* Appellee Br. 55. The Board agreed with Smart Mobile, *Final Written Decision*, 2024 WL 306227, at *10, and so do we. *See id.* at *6–7, 10; J.A. 144 (the petition describing that Ahopelto’s routing functionality “connects data packet networks to each other”). In any event, Apple fares no better with its position, considering that (1) Ahopelto makes no distinctions between “how to forward” and “where to forward” and (2) Dr. Jensen testified that “deciding where to send a packet is an essential part of routing.” *Final Written Decision*, 2024 WL 306227, at *11.

encapsulated data packet.”³ J.A. 833 (Ahopelto), col. 7 ll. 4–5, 32–42; *see also id.*, col. 8 ll. 13–14, 40–42. Because both server and routing functionalities are embedded as a part of Ahopelto’s overarching routing process, the Board reasonably found that Ahopelto’s GGSN does not meet the claim elements.

Apple’s challenge to the Board’s credibility assessment of Dr. Jensen is likewise unpersuasive. “We defer to the Board’s findings concerning the credibility of expert witnesses.” *Yorkey v. Diab*, 601 F.3d 1279, 1284 (Fed. Cir. 2010) (citing *Velandier v. Garner*, 348 F.3d 1359, 1371 (Fed. Cir. 2003)). The Board considered the full record—including Dr. Jensen’s inability to point to any reference that shows implementing the server functionality and the routing functionality as different logical entities. *Final Written Decision*, 2024 WL 306227, at *12. The Board noted that according to Dr. Jensen, “the asserted server functionality includes, or overlaps with, aspects of a ‘GGSN routing process.’” *Id.* at *11 (citing J.A. 709–10 ¶¶ 77, 78). And Smart Mobile’s expert, Dr. Cooklev, opined, with respect to Ahopelto, that “[a skilled artisan] would not understand an alleged ‘server functionality’ to be distinct from an alleged ‘network switch box’ when they perform the same operations.” J.A. 1896–97 ¶ 37. Thus, in view of the presented evidence, the Board reasonably gave “little weight” to Apple’s testimonial evidence because “the opinion fails to show that [a skilled artisan] would have understood that Ahopelto’s GGSN includes a server functionality and a

³ During oral argument, Apple agreed that the Ahopelto’s routing procedure encompasses both functionalities. Oral Arg. at 00:38 – 00:47 (available at https://www.cafc.uscourts.gov/oral-arguments/24-1624_12032025.mp3) (“The prior art discloses multiple steps or functions within its overarching process, and two of those functions disclose the claim elements here.”).

separate and distinct routing functionality.” *Final Written Decision*, 2024 WL 306227, at *12.

Lastly, the Board considered additional cited references and found that those did not alter its conclusion. *Id.* at *13. The Board is correct that Apple did not argue that a skilled artisan would have modified Ahopelto with the teachings of any of the additional references. *Id.* On appeal, Apple relies on the additional references to support Dr. Jensen’s testimony. *See* Appellant Reply Br. 16–17 (arguing that the Board should have given more weight to Dr. Jensen’s testimony because additional references support a skilled artisan’s understanding that “different functions would generally be implemented as ‘different logical (e.g., software) entities.’”). But in view of the entire record, the Board reasonably found that the additional cited references did not alter its obviousness determination.

Based on the foregoing, we affirm the Board’s obviousness determination as to all challenged claims.

CONCLUSION

We find none of Apple’s remaining arguments persuasive. For the foregoing reasons, we *affirm*.

AFFIRMED